

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th May 2014

Subject CT3 – Probability & Mathematical Statistics

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

Q. 1) Following are the sizes of corpus on retirement (in ₹ '000) in a DCS Pension Scheme for the workers of an auto component unit:

208 232 226 219 235 220 234 226 213 220 231 210 218 222 227

- i) Determine the mean and median of these corpus amounts. (2)
 - ii) Determine the lower and upper quartiles, and the interquartile range of these corpus amounts. (3)
- [5]

Q. 2) A large bin contains 3 different types of batteries. The probability that a Type 1 battery will give over 100 hours of use is 0.7, with the corresponding probabilities for Type 2 and Type 3 batteries being 0.4 and 0.3 respectively. Suppose that 20% of the batteries in the bin are of Type 1, 30% of Type 2 and remaining of Type 3.

- i) What is the probability that a randomly chosen battery will give more than 100 hours of use? (2)
 - ii) Given that a battery lasted over 100 hours, what is the conditional probability that it was a Type 1 battery? (2)
- [4]

Q. 3) Compute the missing entries in the following table where X is a random variable with cumulative distribution function $F(x)$:

Distribution	Mean	Variance	x	$F(x)$
Binomial	10	5	*	0.0207
Normal	0	1	*	0.1190
Chi-Square	*	72	51	*

[3]

Q. 4) Let X and Y be two independent random variables.

Define $V = \text{Max}(X, Y)$ and $W = \text{Min}(X, Y)$.

Let F_X , F_Y , F_V and F_W denote the cumulative distribution functions of X , Y , V and W respectively.

- i) Show that $F_V(t) = F_X(t) * F_Y(t)$ (2)
- ii) Show that $F_W(t) = F_X(t) + F_Y(t) - F_X(t) * F_Y(t)$ (2)
- iii) The random variable X has an exponential distribution with mean 0.25 and, independently, Y has an exponential distribution with mean 0.25.

Obtain the distribution function of the minimum of X and Y and derive its median from the first principles. (3)

[7]

- Q. 5)** i) Let X be a random variable which follows a Geometric distribution with parameter p ($0 < p < 1$).

Show that X has a probability generating function:

$$G(s) = \frac{p}{1 - s(1 - p)} \quad \text{for } |s| < \frac{1}{1 - p}$$

Hence find the expected value and the variance of X . (4)

- ii) Consider a pair of random variables X and Y with the following properties:

- X is a discrete random variable
- Y is a continuous random variable
- The unconditional distribution of Y is exponential with parameter λ (> 0)
- Conditional on $Y = y$, X follows a Poisson distribution with expected value y .

- a) Show that the unconditional distribution of X is the distribution described in part (i) for a particular value of p . (3)

- b) Verify that the following results hold:

- $E[X] = E[E\{X|Y\}]$
- $Var[X] = E[Var\{X|Y\}] + Var[E\{X|Y\}]$ (5)

[12]

- Q. 6)** The returns in different years on capital invested by the shareholders in an insurance company are independent and normally distributed with unknown mean (μ) and standard deviation (σ). The Chairman of the Board of Directors claimed in the last Annual General Meeting that he believes that the true values for μ and σ are 21% and 2% respectively.

You are a shareholder of the company and are suspicious about the claim. To verify your suspicion, you considered the reported returns ($R\%$) in the audited accounts over the last 10 years:

20.2 19.3 16.4 18.2 18.1 21.3 26.1 16.8 21.4 20.2

Note: $\sum_{i=1}^{10} r_i = 198.0$; $\sum_{i=1}^{10} (r_i - \bar{r})^2 = 71.08$

- i) Calculate the sample mean and standard deviation for the chosen sample (2)

- ii) Assuming that the Chairman's claim on the true values of μ and σ is correct:

- a) What is the probability that when we pick a new set of 10 years' results, the sample mean will be below the one you calculated in part (i)? (3)

- b) What is the probability that when we pick a new set of 10 years' results, the sample standard deviation will be larger than the one you calculated in part (i)? (3)

[8]

- Q. 7)** i) Suppose X is a continuous random variable with probability density function $f(X)$:

$$f(x) = \begin{cases} \frac{1}{2} x^2 e^{-x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Derive the moment generating function $M_X(t)$. (3)

- ii) The size of a motor insurance claim X , is to be modelled using a gamma random variable with parameters α and θ (both > 0) such that the moment generating function of X is given by:

$$M_X(t) = (1 - \theta t)^{-\alpha} \quad \text{for } t < \frac{1}{\theta}$$

By using only the cumulant generating function of X , show that the coefficient of skewness of X is given by $\frac{2}{\sqrt{\alpha}}$. (3)

[6]

- Q. 8)** A botanist is conducting a crossbreeding experiment with pea plants which resulted in four types of pea pods being grown. He believes that the types are dependent on only two genes – colour and texture, each of which can appear either in dominant or recessive form:

	Colour	Texture
Dominant	Y	R
Recessive	g	w

The actual observed frequencies of the four types YR, Yw, gR, gw in this experiment with 556 plants are as follows:- YR: 315 Yw: 101 gR:108 gw:32

- i) The botanist postulated that the frequencies of the four types YR, Yw, gR, gw should be in the ratio 9:3:3:1. Test whether he is correct in his claim and comment on the result. (4)
- ii) Test if colour and texture are independent traits? (4)

[8]

- Q. 9)** Independent random samples, of size n each, are taken from normal populations $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ respectively, where the parameters μ_1 and μ_2 are unknown and σ^2 is known.

- i) Write down the sampling distributions of $\bar{X}_1$ and $\bar{X}_2$ and hence determine the sampling distribution of $\bar{X}_1 - \bar{X}_2$, the difference between the sample means. (1)
- ii) A statistician wants to conduct a hypothesis test with $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$ at 5% significance level using the following approach:

Step 1: Compute the individual 95% confidence intervals for μ_1 and μ_2 .

Step 2: If the two confidence intervals do not overlap, then reject H_0 .

He asserts that this test will have a probability of Type I error of 0.05. Do you agree? Justify your answer with appropriate statistical reasoning. (6)

[7]

- Q. 10)** The local government has commissioned a study of length of hospital stay, in days, as a function of age, type of health insurance and whether or not the patient died while in the hospital. The length of hospital stay is recorded as a minimum of at least one day and the number of days is rounded up to the next integer.

An analyst working on the study believes that the length of hospital stay (X) can be modelled as a discrete random variable with a probability mass function:

$$P(X = k) = c * \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k = 1, 2, 3 \dots$$

Here, c is a constant and $\lambda (> 0)$ is an unknown parameter.

The analyst collects data for 20 random patients for the study.

- i) Show that:

$$c = \frac{1}{1 - e^{-\lambda}} \quad (2)$$

- ii) Find $l(\lambda)$, the logarithm of the likelihood function of λ based on this sample. (3)

- iii) Find an equation satisfied by $\hat{\lambda}_{MLE}$, the maximum likelihood estimator of λ . (3)
(Do not attempt to solve this equation).

The analyst wrote down the following information obtained using the data on the likelihood equations for various value of λ :

λ_0	1.700	1.750	1.800	1.850
$l(\lambda_0)$	-27.421	-27.391	-27.384	-27.398
$l'(\lambda_0)$	0.824	0.365	-0.072	-0.488
$l''(\lambda_0)$	-9.409	-8.950	-8.527	-8.136

Here:

- $l'(\lambda_0)$ = value of the first derivative of $l(\lambda)$ at $\lambda = \lambda_0$
 - $l''(\lambda_0)$ = value of the second derivative of $l(\lambda)$ at $\lambda = \lambda_0$
- iv) Using linear interpolation and explaining your reasoning carefully, compute an estimate for $\hat{\lambda}_{MLE}$. (4)

- v) It can be shown that the asymptotic variance for $\hat{\lambda}_{MLE}$ is given by:

$$\text{Var}(\hat{\lambda}_{MLE}) \approx - \frac{1}{l'(\hat{\lambda}_{MLE})}$$

- Derive an approximate 95% confidence interval for λ using the information provided. (4)

[16]

- Q. 11)** The following shows how a panel of nutrition experts and a panel of housewives ranked 15 breakfast foods on their palatability:

Breakfast Food	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Nutrition Experts (X)	2	13	14	9	1	4	11	3	7	15	6	12	10	8	5
Housewives (Y)	3	15	13	14	2	6	8	5	4	10	11	9	12	7	1

We are given:

$$\sum x = \sum y = 120; \quad \sum x^2 = \sum y^2 = 1240; \quad \sum \sum xy = 1171$$

- i) Compute sample correlation coefficient. (3)

- ii) Perform an appropriate t-test at the 5% level of significance to check for any linear relationship between the two rankings. (4)

- iii) Compute an approximate 95% confidence interval for the corresponding population parameter ρ . (7)

[14]

- Q. 12)** An Indian market research firm has conducted a survey of five leading mobile brands on various parameters like customer satisfaction, quality of the product and post sales service. Based on the outcome of the survey, it gave scores to each of the brand considering the various parameters. The survey was conducted over six months and following scores were awarded in each month:

Month	Brand				
	A	B	C	D	E
1	168	184	216	192	232
2	176	195	184	184	213
3	141	147	155	181	143
4	166	168	166	194	185
5	136	179	198	149	183
6	133	190	185	189	182
Total	920	1063	1104	1089	1138

$$\sum \sum y_{ij} = 5,314; \quad \sum \sum y_{ij}^2 = 957,718$$

- i) Perform an analysis of variance to investigate whether real differences exist among the scores of all five brands. (6)

- ii) Calculate a 95% confidence interval for the underlying common standard deviation for all five brands assuming there are no significant differences between the mean scores. (4)

[10]
